

Supplementary Activity Game

April 18, 2026

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[3]: import random

class Novice:
    def __init__(self, username):
        self.__username = username
        self.__hp = 100

    def getUsername(self):
        return self.__username

    def getHp(self):
        return self.__hp

    def setHp(self, hp):
        self.__hp = hp

    def basicAttack(self, target):
        damage = random.randint(5, 10)
        target.setHp(target.getHp() - damage)
        print(f"{self.__username} uses Basic Attack on {target.getUsername()}!")

class Swordsman(Novice):
    def slashAttack(self, target):
        damage = random.randint(15, 25)
        target.setHp(target.getHp() - damage)
        print(f"{self.getUsername()} uses Slash Attack! ")

class Archer(Novice):
    def rangedAttack(self, target):
        damage = random.randint(12, 28)
        target.setHp(target.getHp() - damage)
        print(f"{self.getUsername()} uses Ranged Attack! ")

class Magician(Novice):
    def magicAttack(self, target):
        damage = random.randint(20, 35)
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        target.setHp(target.getHp() - damage)
        print(f"{self.getUsername()} casts Magic! ")

class Boss(Novice):
    def __init__(self, username):
        super().__init__(username)
        self.setHp(80) # Boss HP

    def magicAttack(self, target):
        damage = random.randint(10, 20)
        target.setHp(target.getHp() - damage)
        print(f"BOSS {self.getUsername()} uses Dark Magic! ")

def play_match(p1, p2):
    print(f"\nMatch Start: {p1.getUsername()} vs {p2.getUsername()}")
    players = [p1, p2]
    random.shuffle(players)

    while p1.getHp() > 0 and p2.getHp() > 0:
        attacker = players[0]
        defender = players[1]

        if isinstance(attacker, Boss):
            attacker.magicAttack(defender)
        elif isinstance(attacker, Magician):
            attacker.magicAttack(defender)
        elif isinstance(attacker, Swordsman):
            attacker.slashAttack(defender)
        elif isinstance(attacker, Archer):
            attacker.rangedAttack(defender)
        else:
            attacker.basicAttack(defender)

        print(f"{defender.getUsername()} remaining HP: {defender.getHp()}")
        if defender.getHp() <= 0: break
        players.reverse()

    winner = p1 if p1.getHp() > 0 else p2
    print(f"WINNER: {winner.getUsername()}")
    return winner

def game_loop():
    p1_wins = 0
    mode = input("Select Mode (1: Single Player, 2: PvP): ")

    if mode == "1":

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player = Novice("Hero")
while True:
    monster = Boss("Monster")
    winner = play_match(player, monster)
    if winner == player:
        p1_wins += 1
        player.setHp(100 + (p1_wins * 20))
        if p1_wins == 2:
            print("Class Up! Choose: 1-Swordsman, 2-Archer, 3-Magician")
            choice = input()
            if choice == "1": player = Swordsman("Hero")
            elif choice == "2": player = Archer("Hero")
            else: player = Magician("Hero")
        else:
            print(f"Game Over. Total Wins: {p1_wins}")
            break

if __name__ == "__main__":
    game_loop()

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Select Mode (1: Single Player, 2: PvP): 1

Match Start: Hero vs Monster
 Hero uses Basic Attack on Monster!
 Monster remaining HP: 74
 BOSS Monster uses Dark Magic!
 Hero remaining HP: 84
 Hero uses Basic Attack on Monster!
 Monster remaining HP: 68
 BOSS Monster uses Dark Magic!
 Hero remaining HP: 69
 Hero uses Basic Attack on Monster!
 Monster remaining HP: 59
 BOSS Monster uses Dark Magic!
 Hero remaining HP: 53
 Hero uses Basic Attack on Monster!
 Monster remaining HP: 49
 BOSS Monster uses Dark Magic!
 Hero remaining HP: 39
 Hero uses Basic Attack on Monster!
 Monster remaining HP: 40
 BOSS Monster uses Dark Magic!
 Hero remaining HP: 22
 Hero uses Basic Attack on Monster!
 Monster remaining HP: 31
 BOSS Monster uses Dark Magic!

Hero remaining HP: 8
Hero uses Basic Attack on Monster!
Monster remaining HP: 26
BOSS Monster uses Dark Magic!
Hero remaining HP: -2
WINNER: Monster
Game Over. Total Wins: 0

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